Healthcare Facility Provision Across Mumbai's 24 Wards

Data engineering, geospatial analysis and K-Means clustering of listed public health facilities

Academic data-analytics project. Every number in this report is generated from the pipeline outputs (`src/build_report.py`). Generated 2026-10-19.

1. Executive summary

- The MCGM facility layers list **456 public facility records** (dispensaries 186, UPHCs 211, public hospitals 31, maternity homes 28) at **402 distinct sites**. Against a Census 2011 population of 12,442,373, that is **3.23 sites per 100,000 residents** city-wide.
- Ward rates range from 2.17 (R/S) to 5.95 (A) per 100k, a 2.74× spread. The island-city wards (A–G/N) have 4.02 sites per 100k against 2.97 in the suburban wards.
- Facility counts track population closely (Pearson r = 0.829), but more populous wards have *lower* rates (Spearman ρ = -0.572).
- 14 of 24 wards hold a larger share of population than of listed sites. The largest gaps are R/S (+1.82 pp), K/W (+1.79 pp), P/N (+1.35 pp).
- K-Means (K = 4, silhouette 0.316) finds 4 provision profiles. The clustering is exploratory: silhouettes for K = 3–6 are nearly flat, and the grouping depends on the feature set.
- These are **distribution indicators for listed public facilities**. They do not measure capacity, staffing, quality, travel time or private provision, and do not show that any ward is underserved.

2. Problem, questions and scope

The project asks whether listed public healthcare facilities are distributed in proportion to population across Greater Mumbai's 24 administrative wards, and which wards share similar provision profiles. It answers seven questions (Section 7) from verified data only.

3. Data sources, dates and coverage

Dataset	Source	Reference date	Use
Dispensaries, UPHCs, public hospitals, maternity homes (KML)	OpenCity, original source MCGM	resources created 2025-11-25	Facility records, coordinates, source ward
Mumbai Hospitals List (CSV)	OpenCity / MCGM	undated	Reconciliation only (no coordinates)
Mumbai Wards Map (KML)	OpenCity	undated	Ward polygons, area, spatial checks
Ward-wise Census 2011 + Primary Census Abstract	OpenCity / Census of India	2011	Population denominator

Date mismatch. Facilities date from about 2025 and population from 2011. The PRD's 2023 ward population was not supplied. Mumbai's population has grown since 2011, unevenly across wards, so the per-capita rates here are indicative only. Swapping in a newer population table needs one file change (`src/population.py`).

4. Preprocessing, deduplication and ward assignment

- **Ward standardization:** source values such as `KE` and `K/E` map to canonical ids by a structural rule that is verified against the 24 polygon names. Every record mapped.
- **Coordinates:** parsed from KML points; 1 record without coordinates. Most coordinates are rounded to 2 decimals (≈ 1 km), so tolerances are precision-aware.
- **Distinct sites:** records merge only when ward, type-stripped name and coordinates (within 300 m plus rounding error) all agree. 54 pairs matched, mostly a dispensary and a UPHC at the same site, giving 402 sites. No exact or within-layer duplicates were found.
- **Ward assignment:** the source `Ward` attribute is authoritative. A point-in-polygon check agrees for 387 records and is within rounding error for 44 more. 24 records are flagged for manual review.
- **Hospitals CSV:** 18 public hospitals in the CSV are absent from the KML layers: 13 state/central institutions (ESIS, railway, defence and others) and 5 BMC-listed facilities. Counts here cover the MCGM facility layers only.

Details: `docs/data_quality_report.md`.

5. Data quality and integration results

The ward table is driven by the 24-ward master list, so no ward can be dropped by a join. Automated checks confirm that ward totals equal the facility-master totals for every category and for distinct sites, that population and facility shares each sum to 100%, and that the census aggregation reproduces the Primary Census Abstract exactly.

6. Indicator definitions

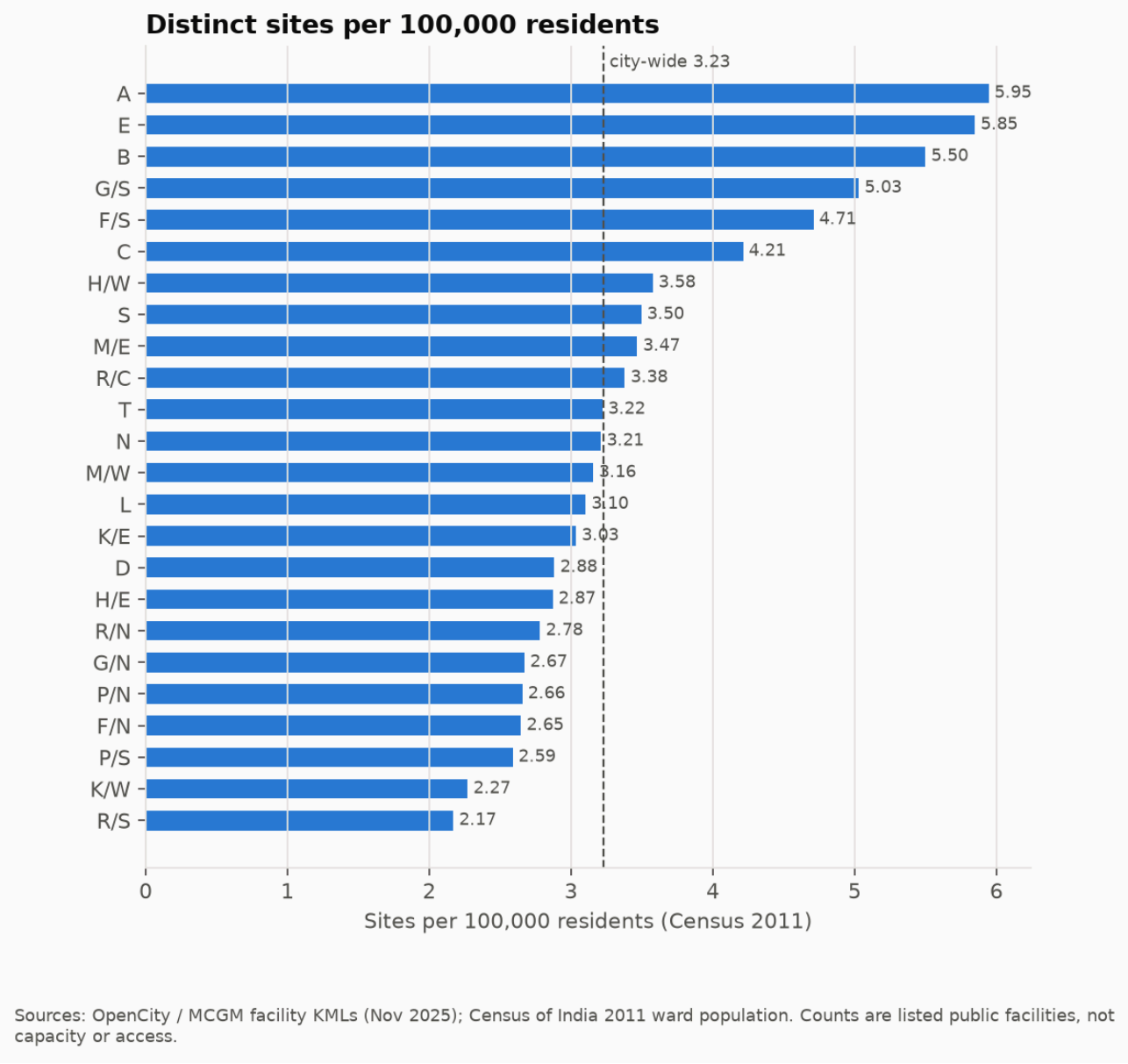
Indicator	Formula	Undefined when
Sites per 100k	distinct sites / population × 100,000	population missing or ≤ 0
Population per site	population / distinct sites	zero sites
Category rate per 100k	category records / population × 100,000	population missing or ≤ 0
Category share of sites	category records / distinct sites × 100 (can sum > 100%: co-located sites)	zero sites
Population share	ward population / total × 100	—
Facility share	ward sites / total sites × 100	—
Share gap	population share − facility share (pp; positive = relatively fewer listed sites)	—

Undefined values stay empty (NaN); they are never filled with zero. All 24 wards have at least one site and a positive population, so no indicator is undefined in the real data. The edge cases are covered by unit tests.

Ward	Pop. 2011	Disp.	UPHC	Hosp.	Mat.	Sites	Sites/100k	Pop./site	Gap (pp)	Cluster
A	185,014	6	3	4	0	11	5.95	16,819	-1.25	4
B	127,290	5	2	0	0	7	5.50	18,184	-0.72	3
C	166,161	5	3	0	1	7	4.21	23,737	-0.41	3
D	346,866	6	5	0	0	10	2.88	34,687	0.30	1
E	393,286	13	7	5	0	23	5.85	17,099	-2.56	4
F/S	360,972	9	7	2	1	17	4.71	21,234	-1.33	2
F/N	529,034	7	8	2	1	14	2.65	37,788	0.77	1
G/S	377,749	13	8	0	1	19	5.03	19,882	-1.69	3
G/N	599,039	9	9	0	1	16	2.67	37,440	0.83	1

Ward	Pop. 2011	Disp.	UPHC	Hosp.	Mat.	Sites	Sites/100k	Pop./site	Gap (pp)	Cluster
H/E	557,239	7	9	1	1	16	2.87	34,827	0.50	1
H/W	307,581	5	6	1	0	11	3.58	27,962	-0.26	2
K/E	823,885	11	12	1	3	25	3.03	32,955	0.40	1
K/W	748,688	7	10	1	1	17	2.27	44,040	1.79	1
P/S	463,507	3	8	1	1	12	2.59	38,626	0.74	2
P/N	941,366	11	13	2	3	25	2.66	37,655	1.35	1
R/S	691,229	6	10	1	1	15	2.17	46,082	1.82	1
R/C	562,162	7	10	1	3	19	3.38	29,587	-0.21	2
R/N	431,368	5	8	1	0	12	2.78	35,947	0.48	2
L	902,225	15	17	1	2	28	3.10	32,222	0.29	2
M/E	807,720	11	15	1	2	28	3.47	28,847	-0.47	2
M/W	411,893	6	8	1	1	13	3.16	31,684	0.08	2
N	622,853	8	11	2	1	20	3.21	31,143	0.03	2
S	743,783	8	15	1	3	26	3.50	28,607	-0.49	2
T	341,463	3	7	2	1	11	3.22	31,042	0.01	2

7. Exploratory findings



Sources: OpenCity / MCGM facility KMLs (Nov 2025); Census of India 2011 ward population. Counts are listed public facilities, not capacity or access.

Figure 1. Distinct sites per 100,000 residents (Census 2011).

Q1 – Highest and lowest counts. Most sites: L (28), M/E (28), S (26). Fewest: D (10), B (7), C (7). Raw counts mostly follow population (r = 0.829), so ranking by count favours large wards.

Q2 – Lowest per-capita provision. Lowest rates: R/S (2.17), K/W (2.27), P/S (2.59), F/N (2.65) per 100k. Highest: A (5.95), E (5.85), B (5.50). *Limitation:* the denominators are 2011 populations. Fast-growing suburban wards are likely under-counted, which would make their true rates lower still.

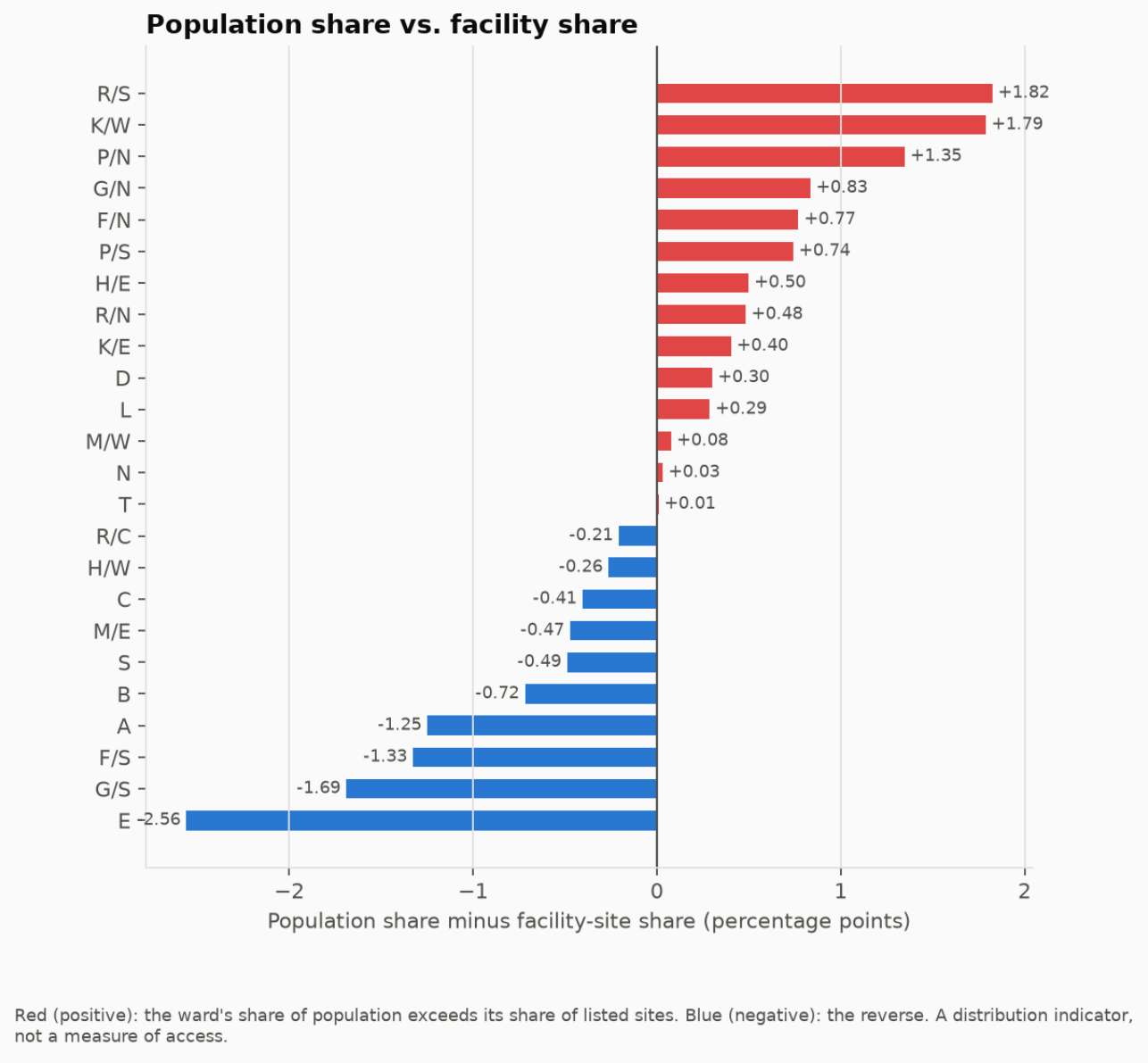


Figure 2. Population share minus facility-site share, percentage points.

Q3 – Population share vs facility share. 14 wards have a positive gap. The largest: R/S (+1.82 pp), K/W (+1.79 pp), P/N (+1.35 pp), G/N (+0.83 pp). The most negative: E (-2.56 pp), G/S (-1.69 pp), F/S (-1.33 pp).

Q4 – Facility mix. Public hospitals are concentrated: E (5) and A (4) hold 9 of 31 hospital records, which include the medical-college hospitals. Wards with no listed public hospital: B, C, D, G/S, G/N. Wards with no listed maternity home: A, B, D, E, H/W, R/N. Dispensary rates are highest in the island city (B 3.93, G/S 3.44, E 3.31, A 3.24). UPHC rates vary much less across wards (1.34–2.12 per 100k). IQR outliers are listed in `outputs/tables/eda_iqr_outliers.csv`.

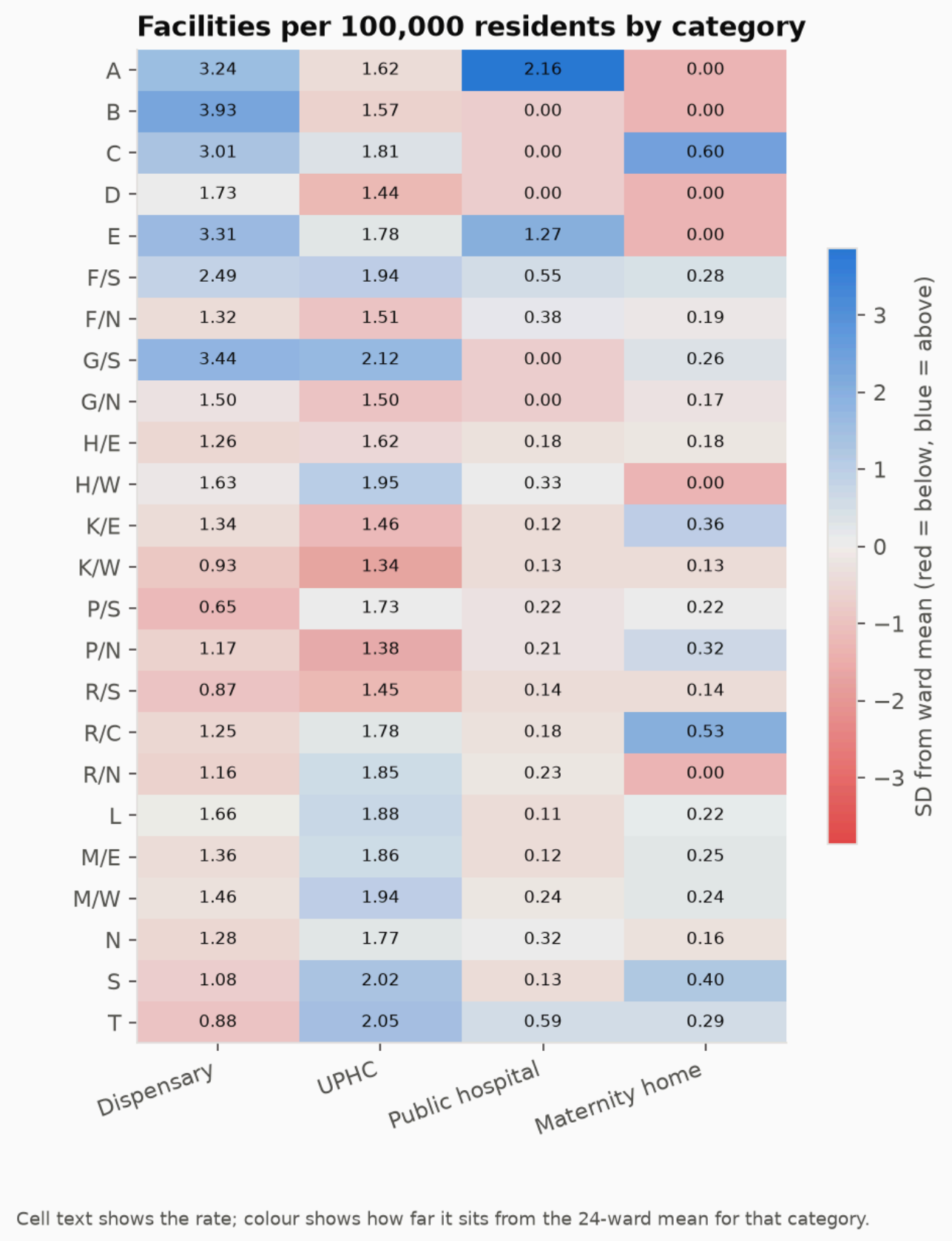


Figure 3. Category rates per 100k; colour = SD from the 24-ward mean (red below, blue above).

Q5 – Geographic pattern. Higher rates cluster in the southern island city, which is the historic core with the major teaching hospitals. Lower rates appear in the large western and eastern suburbs. Because most coordinates are rounded to about 1 km, the map is valid at ward scale, not street scale.

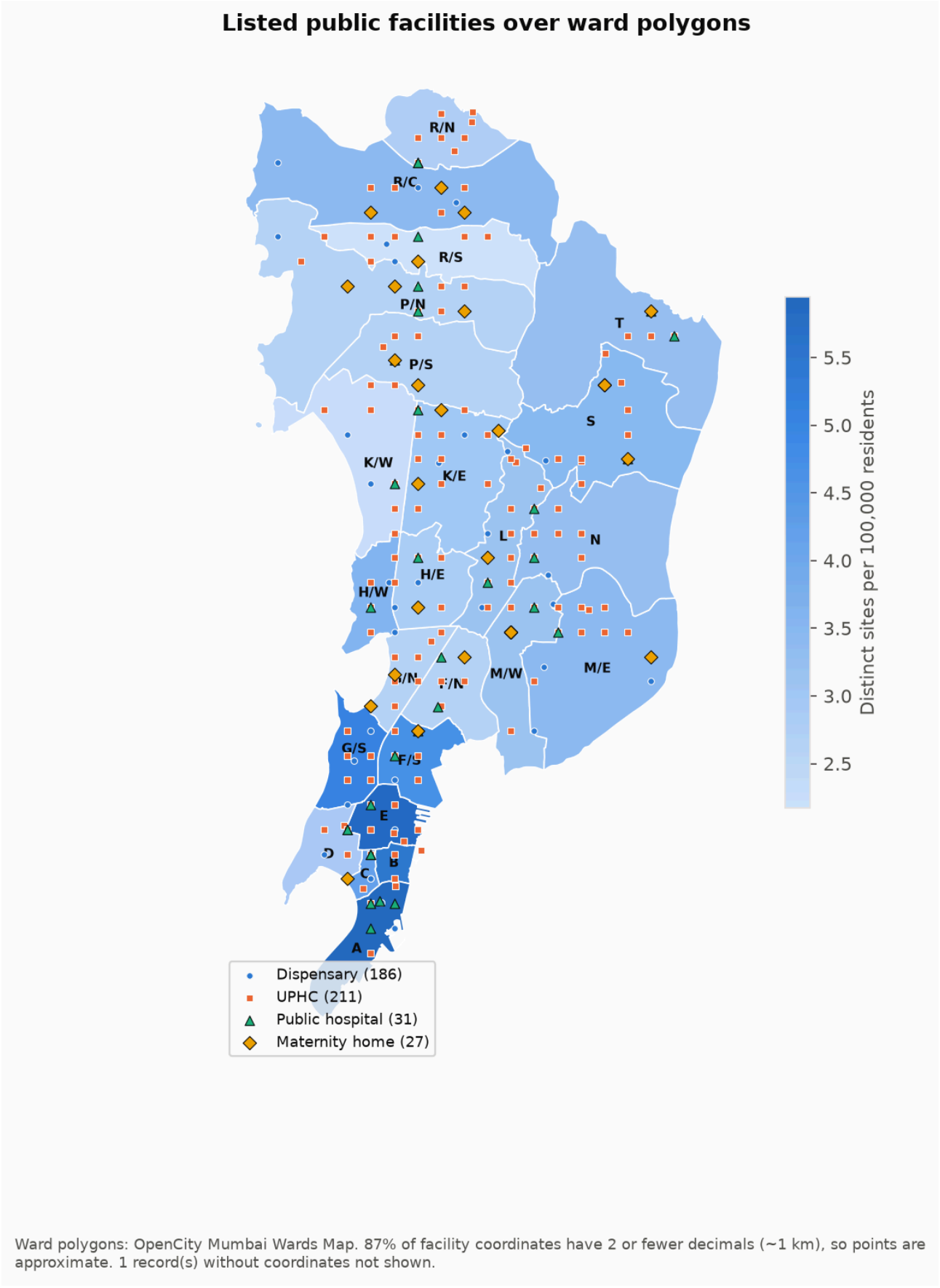


Figure 4. Facility locations over ward polygons shaded by sites per 100k.

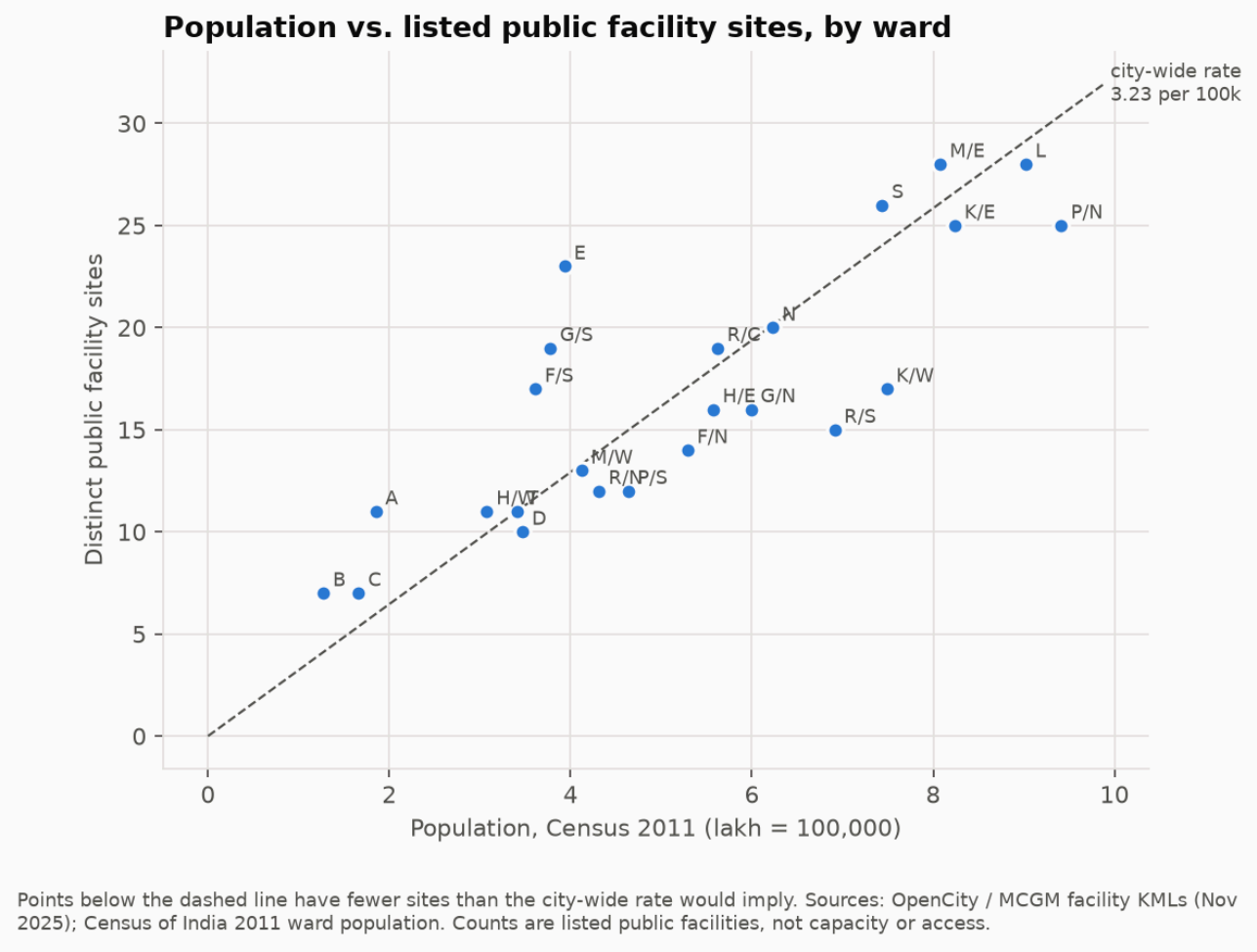


Figure 5. Ward population vs distinct sites; dashed line = city-wide rate.

8. Machine learning: K-Means clustering

Objective. Group wards by similarity of provision profile. No independently defined target (such as measured access or outcomes) exists, so supervised learning is not justified. A label derived from these same rates would be circular.

Features. Rates per 100k for the four categories: dispensaries, UPHCs, public hospitals and maternity homes. Their pairwise Spearman correlations are weak ($|\rho| \leq 0.29$), so they are not redundant. The overall site rate and population-per-site are excluded because they are functions of these rates, and population is excluded because it already sits in every denominator. log1p was applied before z-scoring to damp the A/E hospital outliers. Configuration: StandardScaler → KMeans, n_init = 50, max_iter = 500, random_state = 42, scikit-learn 1.9.1.

Choosing K. Every candidate K was fitted and checked for silhouette, inertia, cluster sizes and stability (30 seeds, 100 × 80% subsamples, 100 × 5% multiplicative noise; adjusted Rand index against the full-data solution):

k	inertia	silhouette	cluster_sizes	status	seed_ari_min	subsample80_ari_mean	subsample80_ari_p10	noise5pct_ari_mean
2	68.343	0.456	[22, 2]	ok	1.000	0.599	-0.029	0.963
3	46.184	0.303	[13, 9, 2]	ok	1.000	0.685	0.369	0.673
4	35.015	0.316	[11, 8, 3, 2]	ok	0.888	0.727	0.337	0.640
5	27.351	0.315	[10, 7, 3, 2, 2]	ok	0.781	0.752	0.512	0.630

k	inertia	silhouette	cluster_sizes	status	seed_ari_min	subsample80_ari_mean	subsample80_ari_p10	noise5pct_ari_mean
6	21.353	0.327	[8, 7, 3, 2, 2, 2]	ok	0.747	0.778	0.559	0.662

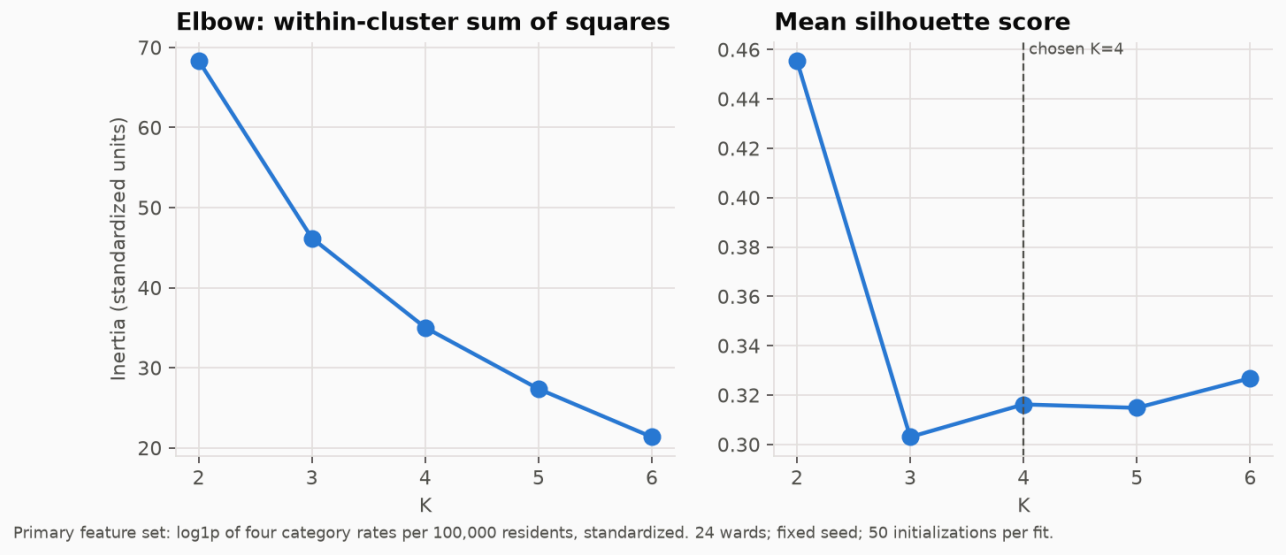


Figure 6. Elbow and silhouette diagnostics.

K = 2 has the highest silhouette (0.456), but it only separates two outlier wards (A, E) from the rest. It is also unstable under subsampling: its 10th-percentile ARI is -0.03, because the two-ward cluster disappears whenever A or E is left out. There is no clear elbow, and silhouettes for K = 3–6 fall within 0.024 of each other. **K = 4 was chosen as a documented compromise:** 1. It keeps the A/E split. 2. It splits the other 22 wards into the same three groups that an independent K = 3 run on those 22 wards produces. 3. It has a single two-ward cluster, against two at K = 5 and three at K = 6. 4. Its stability is moderate: seed-ARI minimum 0.89, mean subsample ARI 0.73, mean noise ARI 0.64.

The choice involves judgement and is not an objective optimum. The K = 2 labels are exported alongside the K = 4 labels.

Cluster profiles. Clusters are numbered by ascending mean sites per 100k. Profile wording compares each cluster with the 24-ward mean (±0.5 SD) and is relative to this dataset only.

Cluster	n	Wards	Profile	Silh.	Disp./100k	UPHC/100k	Hosp./100k	Mat./100k	Sites/100k
1	8	D, F/N, G/N, H/E, K/E, K/W, P/N, R/S	Relatively lower UPHC & dispensary rates	0.43	1.26	1.46	0.15	0.19	2.65
2	11	F/S, H/W, P/S, R/C, R/N, L, M/E, M/W, N, S, T	Relatively higher UPHC rates	0.26	1.35	1.89	0.28	0.24	3.34
3	3	B, C, G/S	Relatively higher dispensary rates; lower public-hospital rates	0.00	3.46	1.83	0.00	0.29	4.91
4	2	A, E	Relatively higher public-hospital & dispensary rates; lower maternity-home rates	0.64	3.27	1.70	1.72	0.00	5.90

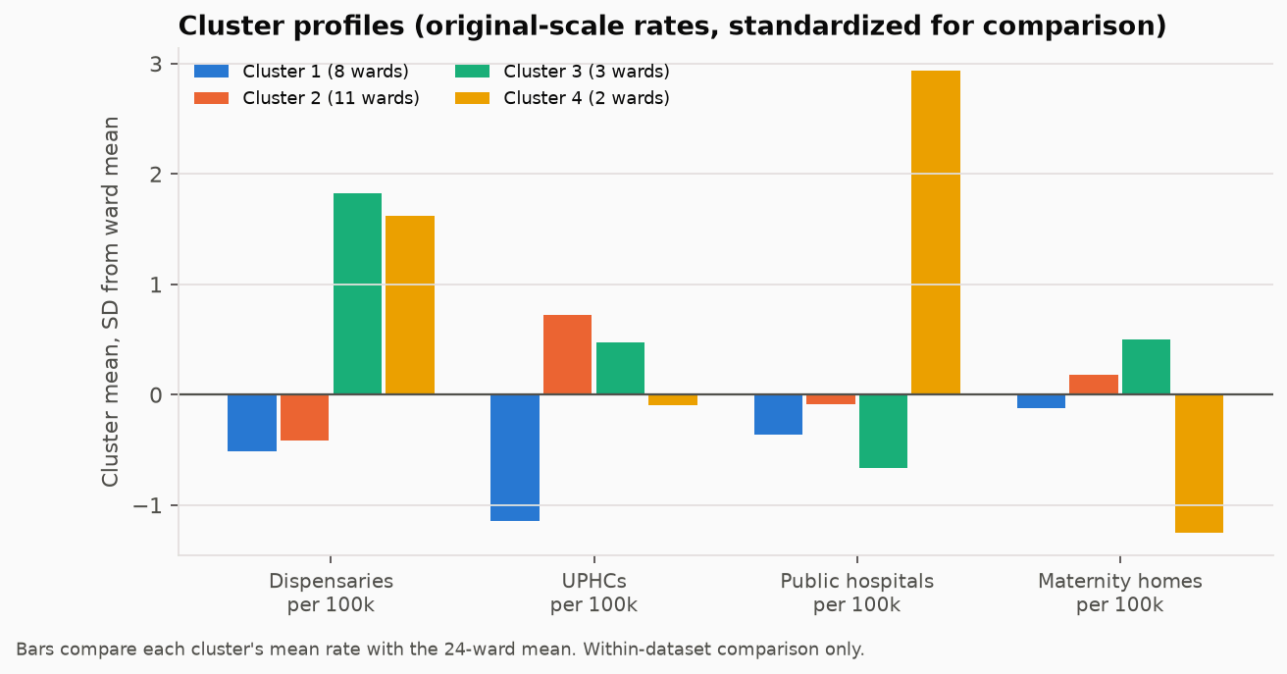


Figure 7. Cluster mean of each feature in SD units from the ward mean.

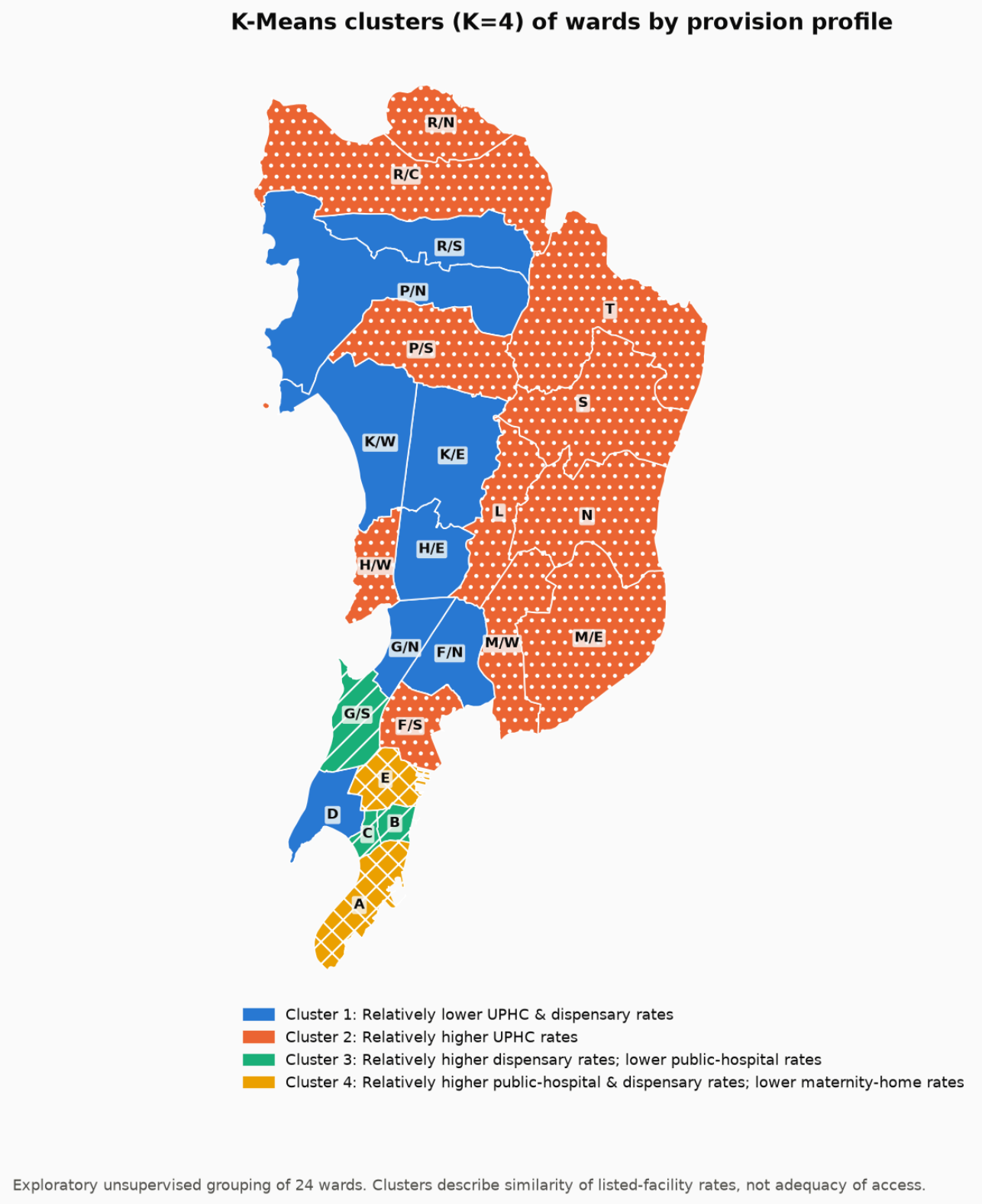


Figure 8. Ward-to-cluster map.

Sensitivity to the feature set. ARI against the final clustering at K = 4: primary_log_category_rates 1.00; raw_category_rates 0.89; primary_vs_secondary_care 0.24; overall_rate_and_density 0.37. Removing the log transform changes little. Coarser feature sets (primary vs secondary care; overall rate + density) produce substantially different groupings. The clusters reflect *which* provision dimensions are emphasised, not a unique natural partition.

Limitations. There are only 24 observations; cluster 3 has a mean silhouette near zero, so its wards sit between clusters; and the model has no external validation. Internal metrics say nothing about predictive or

clinical validity.

9. Power BI report

Direct authoring is blocked because Power BI Desktop runs on Windows only and this project was built on macOS. No PBIX was created or tested. Import-ready tables (`powerbi/data/`), the relationship design, DAX measures, a three-page layout and step-by-step setup are in `powerbi/powerbi_spec.md`. `powerbi/data/expected_kpis.json` holds the KPI values the finished report must reproduce.

10. Findings and recommendations for further investigation

Wards that combine a low site rate with a positive share gap: **F/N, K/W, P/S, R/S**. Cluster 1 (lower dispensary and UPHC rates) contains D, F/N, G/N, H/E, K/E, K/W, P/N, R/S. These wards merit further investigation. That does not mean they are underserved: their absolute site counts are often high, and residents may use facilities in neighbouring wards or private providers.

Data needed to judge access: 1. Current (2023+) ward population estimates. 2. Beds, OPD capacity and staffing per facility. 3. Travel time on the road and transit network rather than ward membership. 4. Private and state/central facilities, starting with the 18 public hospitals found only in the CSV. 5. Utilisation and health-outcome data. 6. Exact facility coordinates.

11. Limitations, conclusion and reproducibility

Limitations: * 2011 population against 2025 facilities. * Facility counts carry no weight for size or capacity. * Some public hospitals are missing from the layers. * Coordinates are rounded. * Co-location is a rule-based judgement. * Ward polygons are undated. * Only 24 observations for clustering.

Conclusion. Listed public facilities follow population in absolute terms. Per resident, the island-city wards have 1.35× the suburban rate (4.02 vs 2.97 sites per 100k). The K-Means grouping summarises four provision profiles, but it is exploratory and sensitive to the feature set.

Reproduce:

```
python3 -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/python run_pipeline.py
.venv/bin/python -m pytest -q
.venv/bin/python -m src.build_report
```

References. * OpenCity – Mumbai City Public Health Centres (data.opencity.in/dataset/mumbai-city-public-health-centres-maps). * OpenCity – Mumbai Wards Map (data.opencity.in/dataset/mumbai-wards-map). * OpenCity – Mumbai Ward-wise Census Data (data.opencity.in/dataset/mumbai-ward-wise-census-data). * Census of India 2011, Primary Census Abstract. * Pedregosa et al., *Scikit-learn: Machine Learning in Python*, JMLR 12 (2011). * Rousseeuw, *Silhouettes*, J. Comput. Appl. Math. 20 (1987). * Hubert & Arabie, *Comparing partitions*, J. Classification 2 (1985).